CS 306: Physical Layer (Part – I)

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Outline

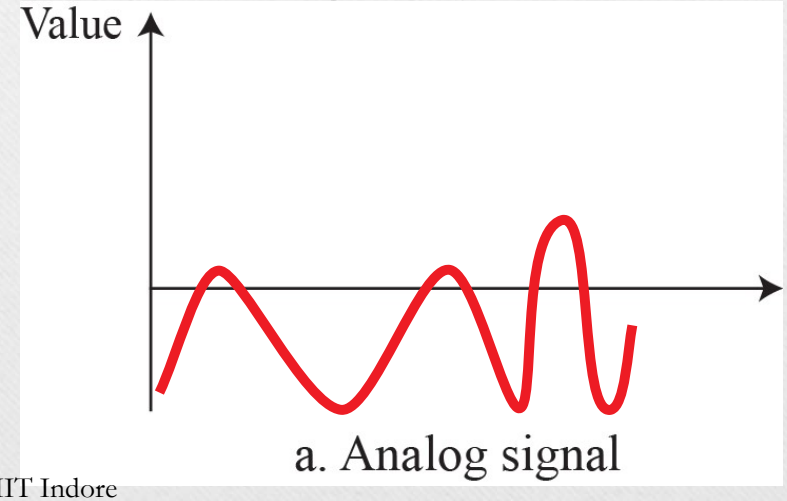
- DATA AND SIGNALS
- PERIODIC ANALOG SIGNALS
- DIGITAL SIGNALS

Data

- Analog data refers to information that is continuous.
- Digital data refers to information that has discrete states.

Analog Signal

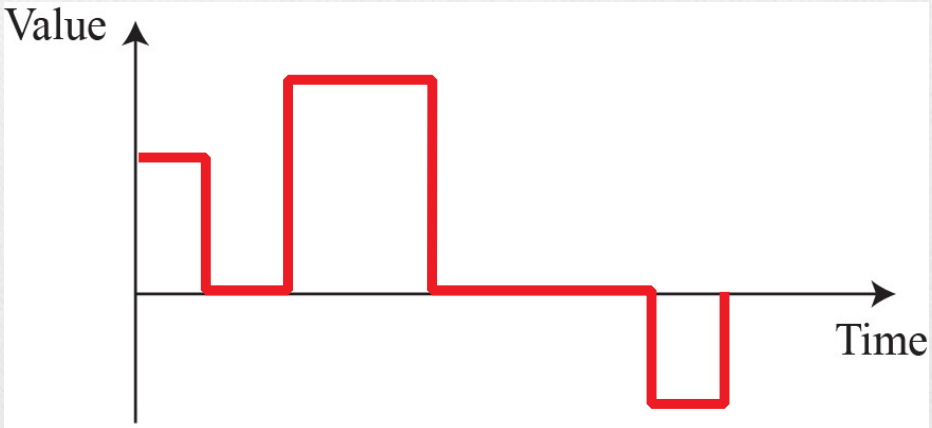
- An analog signal has infinitely many levels of intensity over time. As the wave moves from value A to value B, it passes through and includes an infinite number of values along its path.



a. Analog signal

Digital Signal

- A digital signal, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0.



b. Digital signal

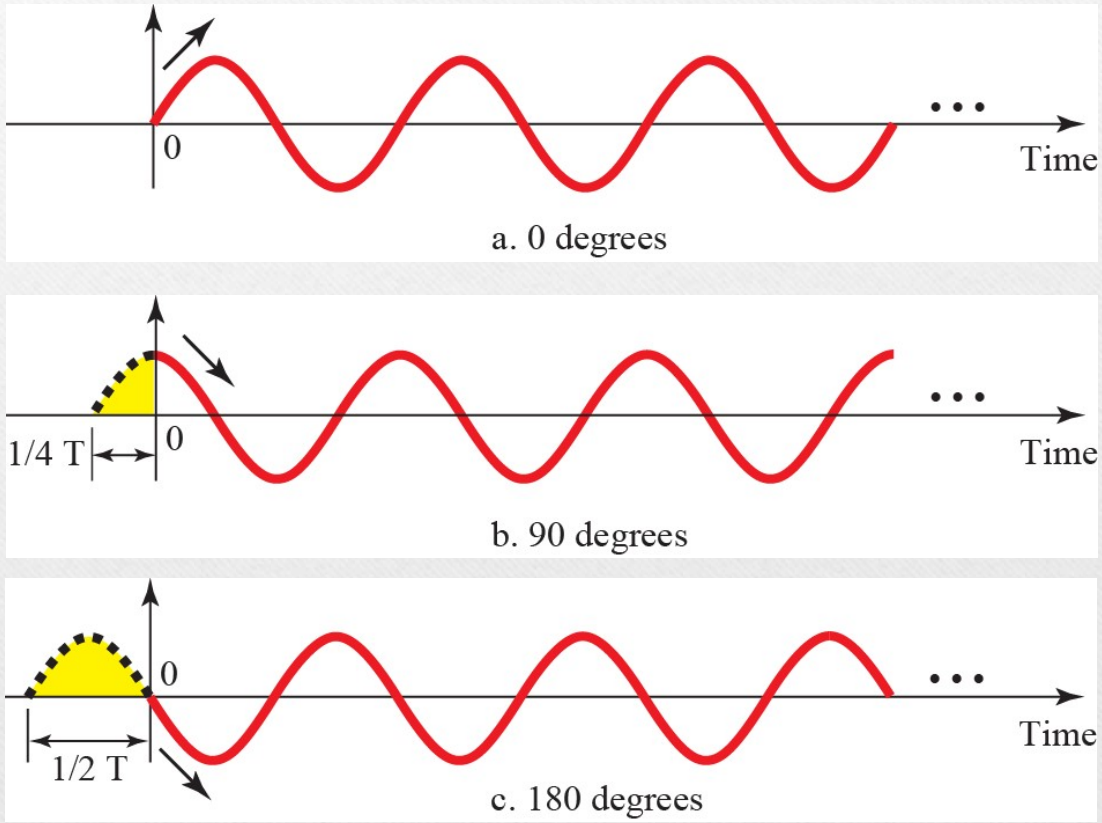
Periodic and Nonperiodic

- A periodic signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods.
- The completion of one full pattern is called a cycle.
- A nonperiodic signal changes without exhibiting a pattern or cycle that repeats over time. Both analog and digital signals.

Phase

- The term phase, or phase shift, describes the position of the waveform relative to time 0.
- If we think of the wave as something that can be shifted backward or forward along the time axis, phase describes the amount of that shift. It indicates the status of the first cycle.

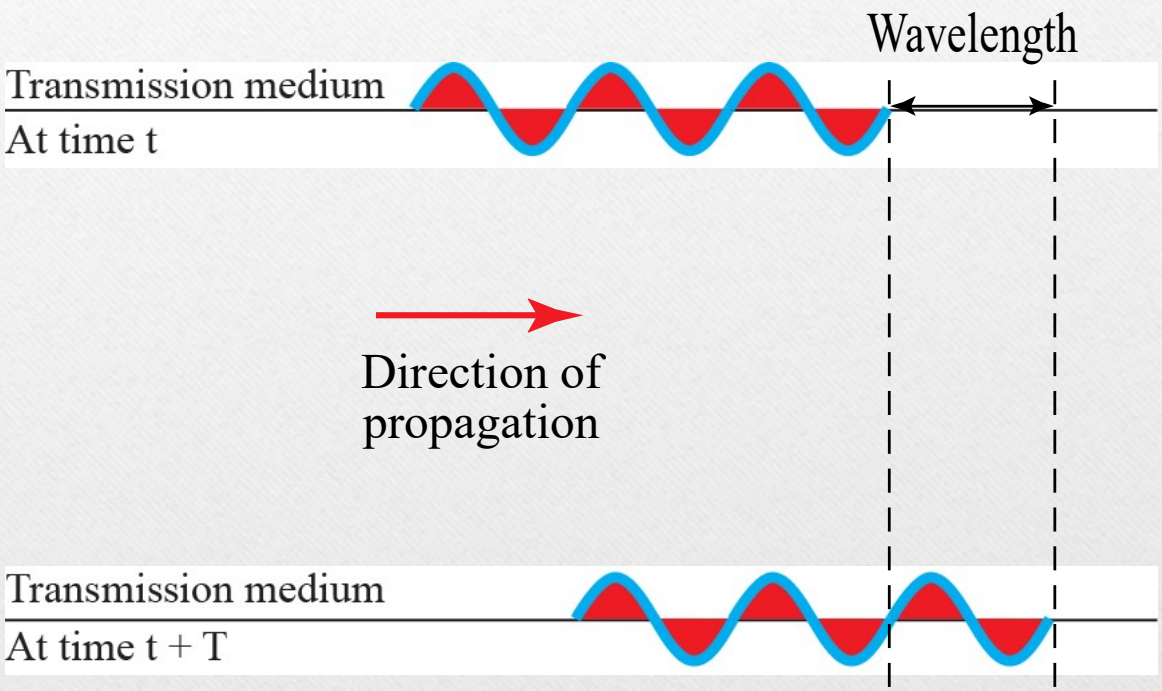
Phase (Contd.)


0
Time
a. 0 degrees
1/4 T
0
Time
b. 90 degrees
0
1/2 T
Time
c. 180 degrees

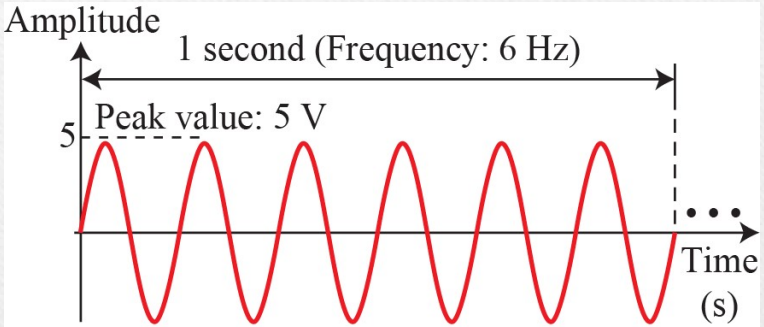
Wavelength

- Wavelength is another characteristic of a signal traveling through a transmission medium. Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium.

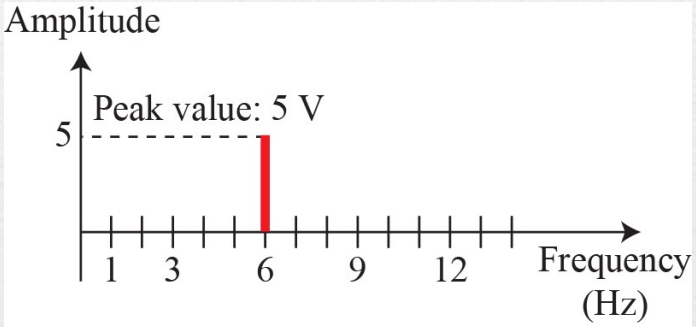
Wavelength (Contd.)



Time and Frequency Domains

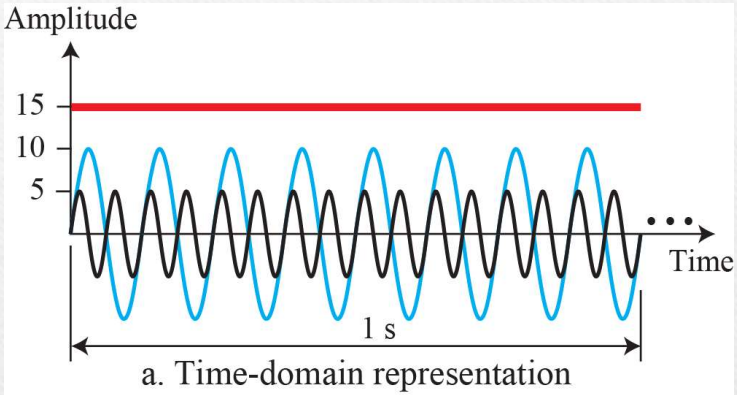


a. A sine wave in the time domain

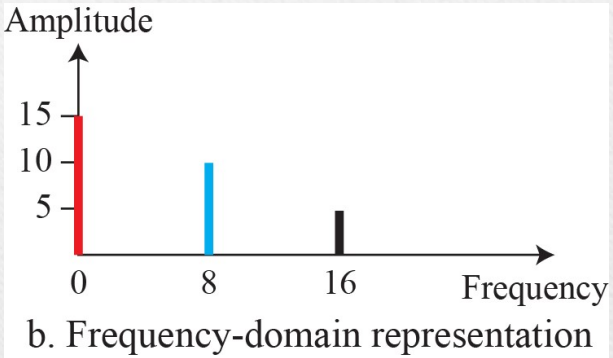


b. The same sine wave in the frequency domain

Time and Frequency Domains (Contd.)

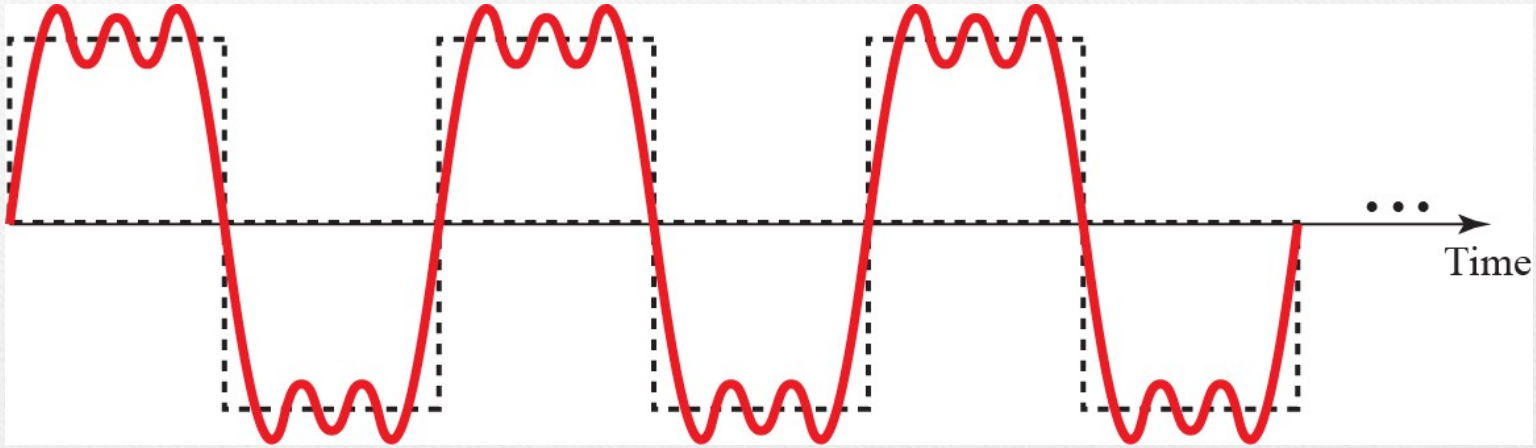


a. Time-domain representation

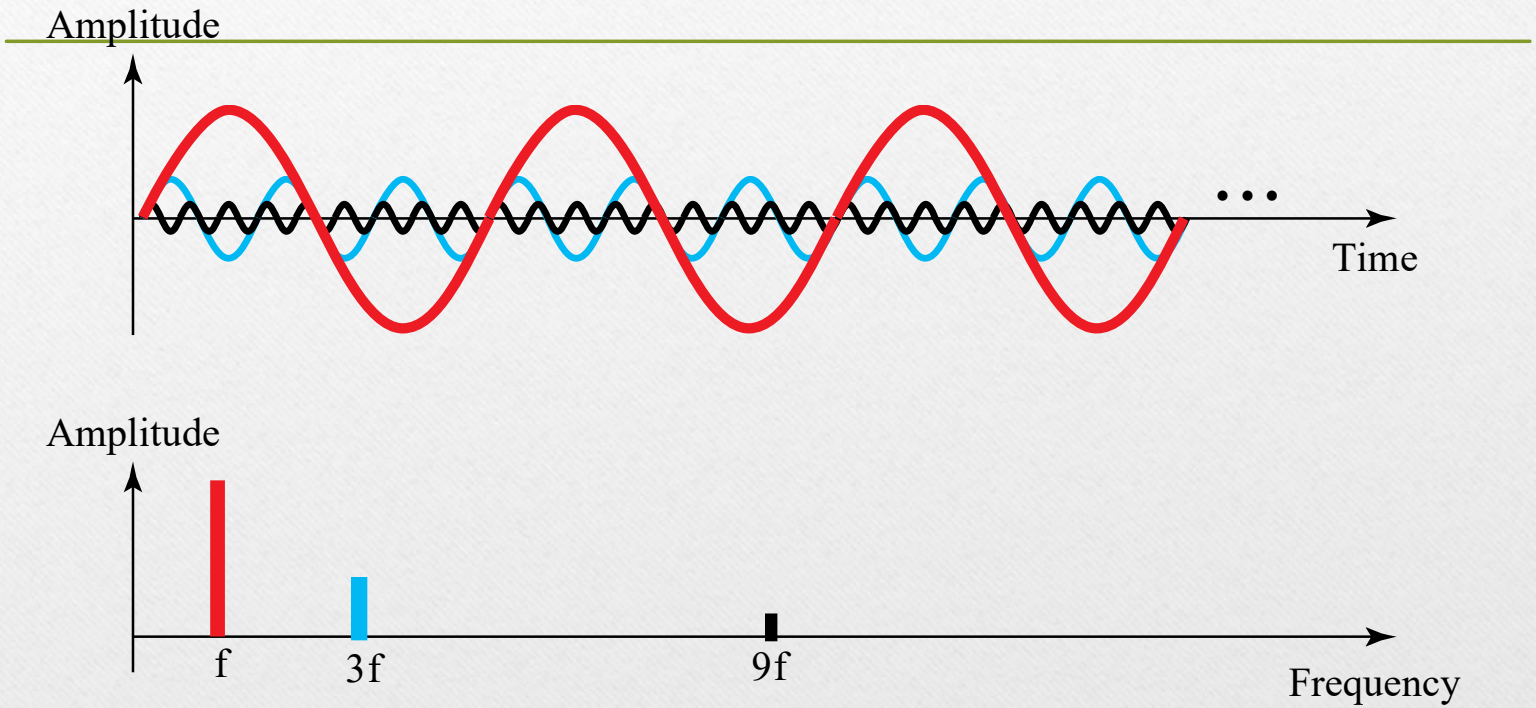


b. Frequency-domain representation

Composite Signals



Composite Signals

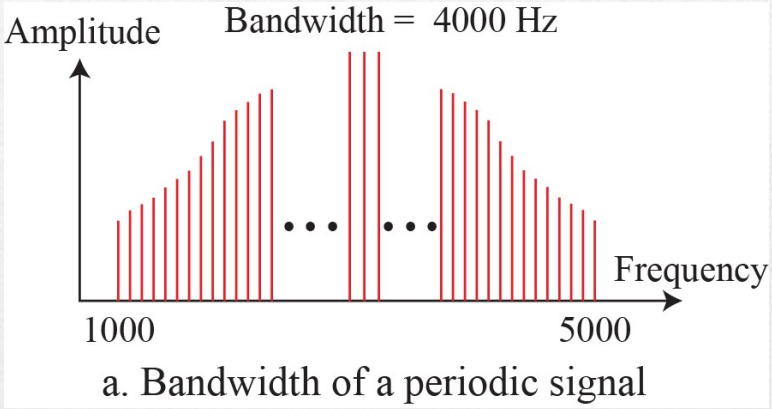


b. Frequency-domain decomposition of the composite signal

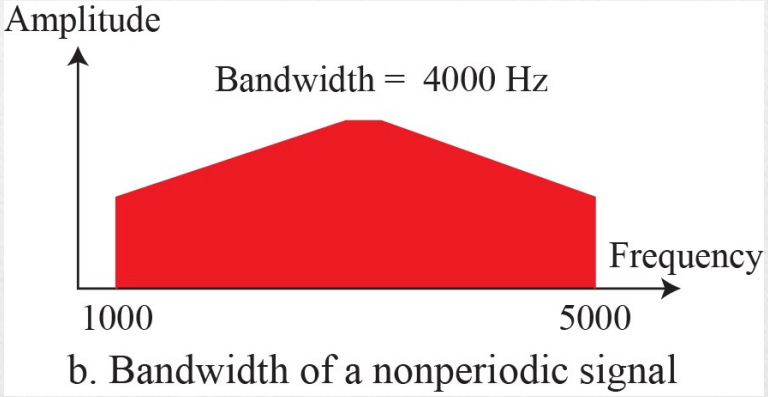
Bandwidth

- The range of frequencies contained in a composite signal is its bandwidth. The bandwidth is normally a difference between two numbers.
- For example, if a composite signal contains frequencies between 1000 and 5000, its bandwidth is 5000 − 1000, or 4000.

Bandwidth (Contd.)

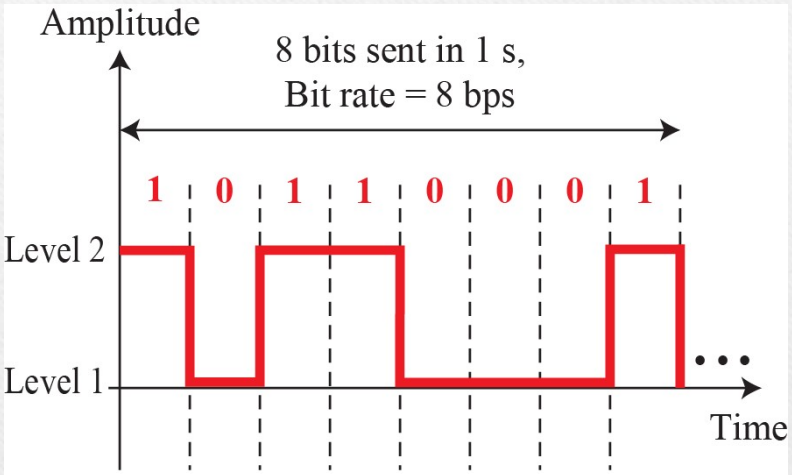


a. Bandwidth of a periodic signal

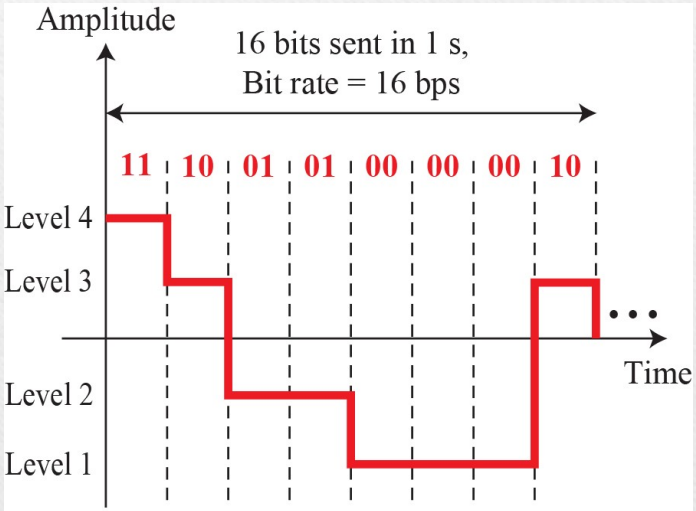


b. Bandwidth of a nonperiodic signal

Digital Signal



a. A digital signal with two levels



b. A digital signal with four levels

Bit Rate and Bit Length

- The bit rate is the number of bits sent in 1s, expressed in bits per second (bps).
- The bit length is the distance one bit occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

Transmission Impairment

- Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium
- Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies.

Transmission Impairment (Contd.)

- Noise is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- Thermal noise is the random motion of electrons in a wire, which creates an extra signal not originally sent by the transmitter.
- Induced noise comes from sources such as motors.
- Crosstalk is the effect of one wire on the other.

Data Rate Limit

- **Nyquist Rate:** For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate.

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

- **Shannon Capacity:** The Shannon capacity determines the theoretical highest data rate for a noisy channel.

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

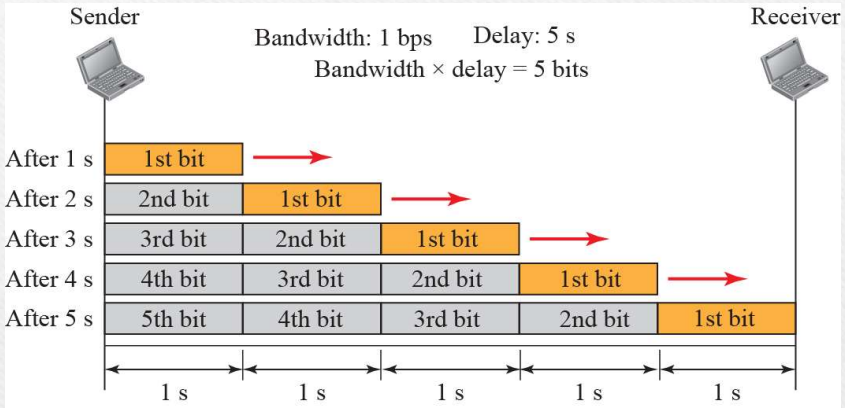
Performance

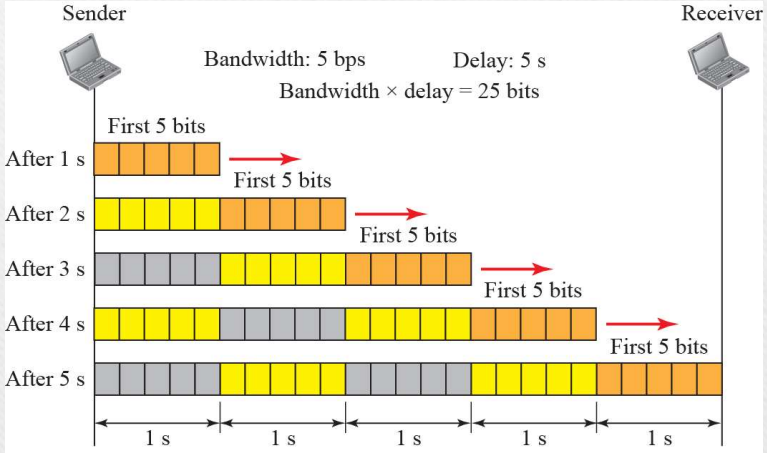
- **Bandwidth:** For performance, we evaluate the bandwidth in terms of maximum bps that can be achieved
- **Throughput:** The throughput is a measure of how fast we can send data through a network.
- **Latency:** The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

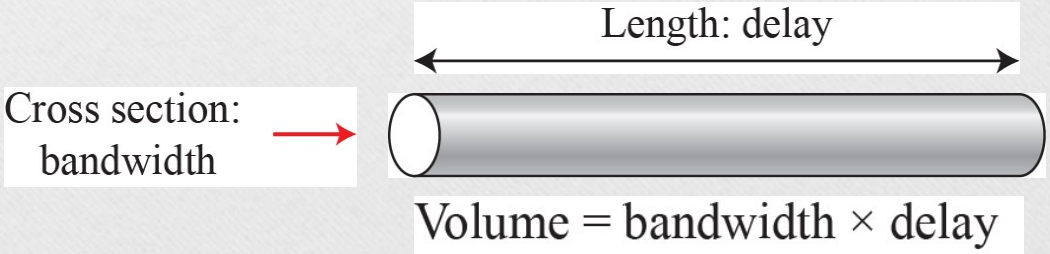
Latency = propagation time + transmission time + queuing time + processing delay

Performance (Contd.)

- **Bandwidth-delay product:**







Performance (Contd.)

- **Jitter:** Jitter is a problem where different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive.

Next Class

- Digital Transmission

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